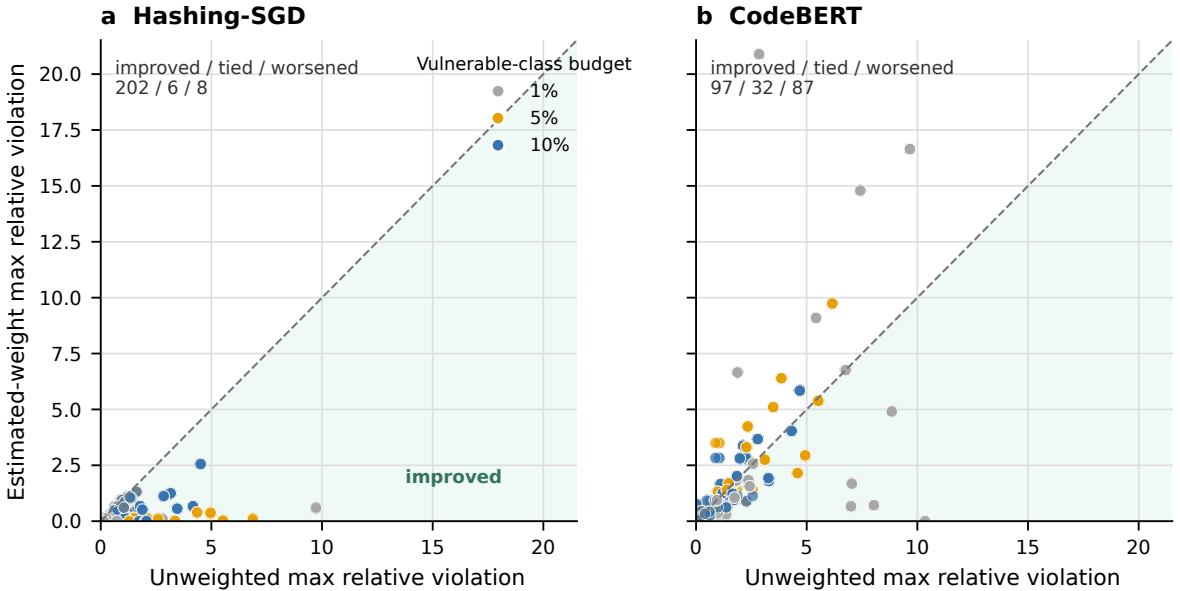Hashing improves risk alignment across the frozen operating grid